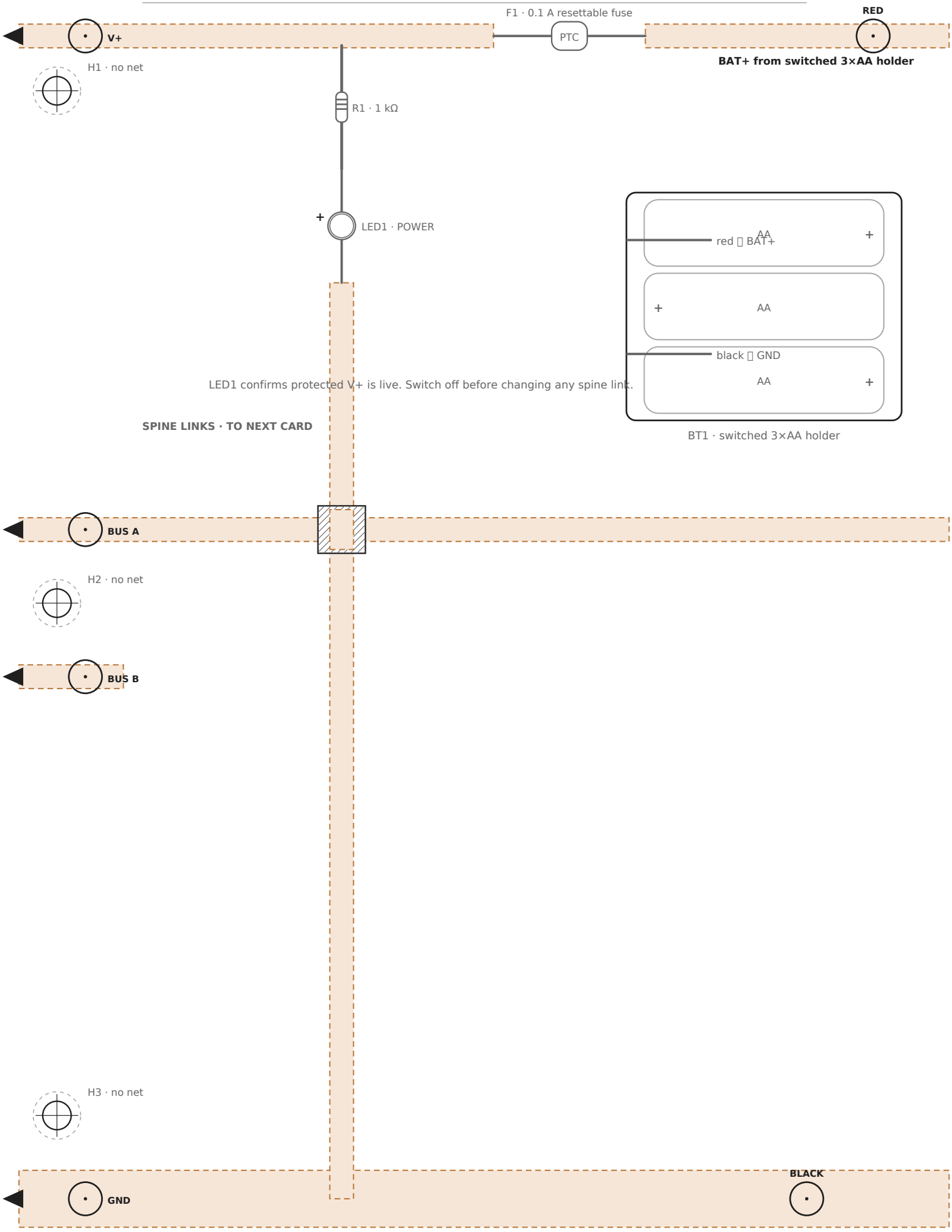
Card 1A · fused power entry

T6 · 1/10

A switched 3×AA holder and resettable fuse feed the book from its first page.

V2 four-card synth test



Print at 100 %: this bar should measure 100 mm.

T6 Card 1: power back and book map

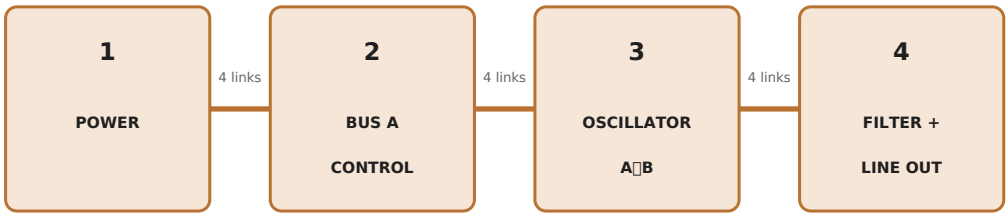
Follow the four-card signal flow before laying copper or joining the spine.

V+ B

V+

H1 · no net

T6 BOOK CONNECTION MAP



SPINE LINKS · TO NEXT CARD

V+ · BUS A · BUS B · GND at every card boundary

BUS A · spine stub

BUS A

H2 · no net

BUS B · work rail

BUS B

Each boundary gets four independent flexible links. V+ and GND cross every card; BUS A is used on side A and BUS B on side B. The rings remain mechanical only.

H3 · no net

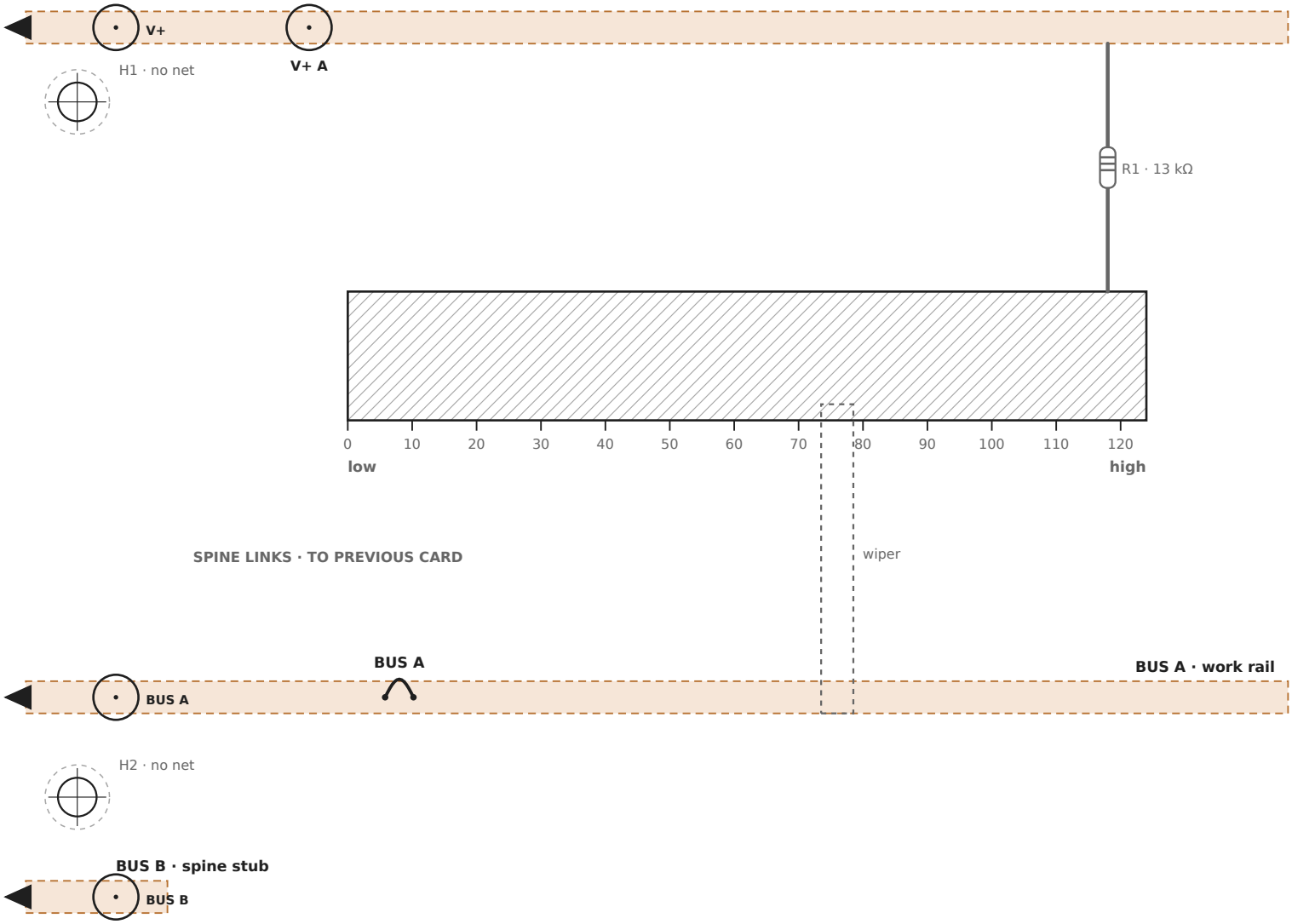
GND B

GND

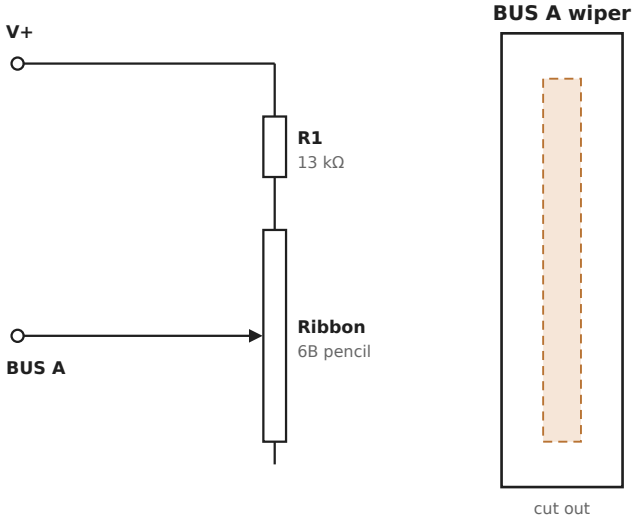
Print at 100 %: this bar should measure 100 mm.

Card 2A · BUS A pencil controller

The controller creates pitch control on side A and sends it through the book.



Controller schematic



Card 2A · BUS A pencil controller

Adapted from M3. V+ feeds a 13 kΩ resistor and pencil ribbon. The movable wiper drives BUS A; no audio or load current belongs on this bus.



T6 Card 2: controller back

BUS A uses only a short spine stub here; BUS B crosses its named side-B work area.



H1 · no net

Pass-through face

This face contains no synth function. Wrap BUS A only far enough to reach its spine link, and carry BUS B across side B. Do not extend BUS A through this work area.

SPINE LINKS · TO NEXT CARD

BUS A · spine stub



H2 · no net

BUS B · work rail

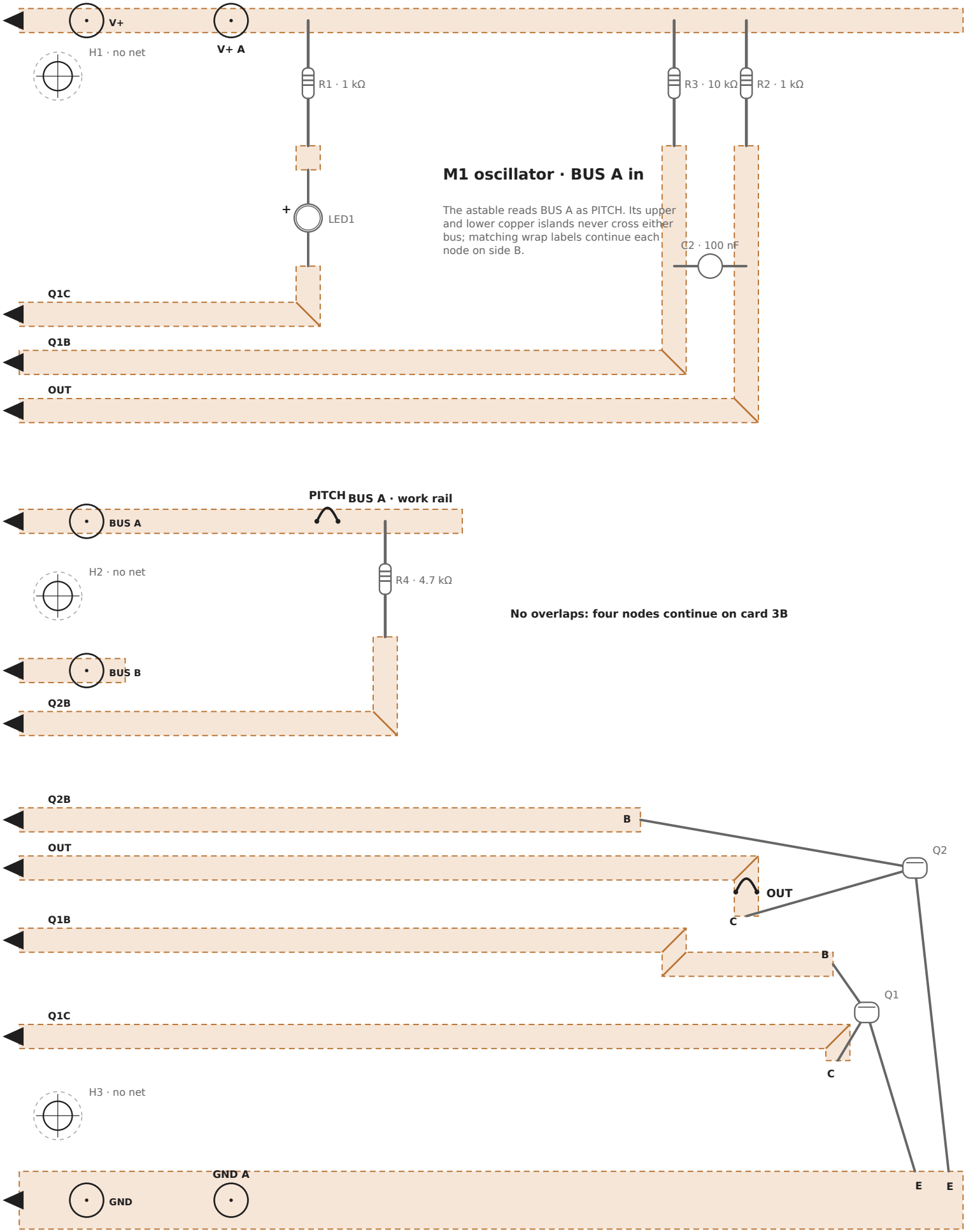


H3 · no net



T6 Card 3: oscillator input

BUS A ends at R4; four separated node pairs wrap to the reverse routing layer.



Print at 100 %: this bar should measure 100 mm.

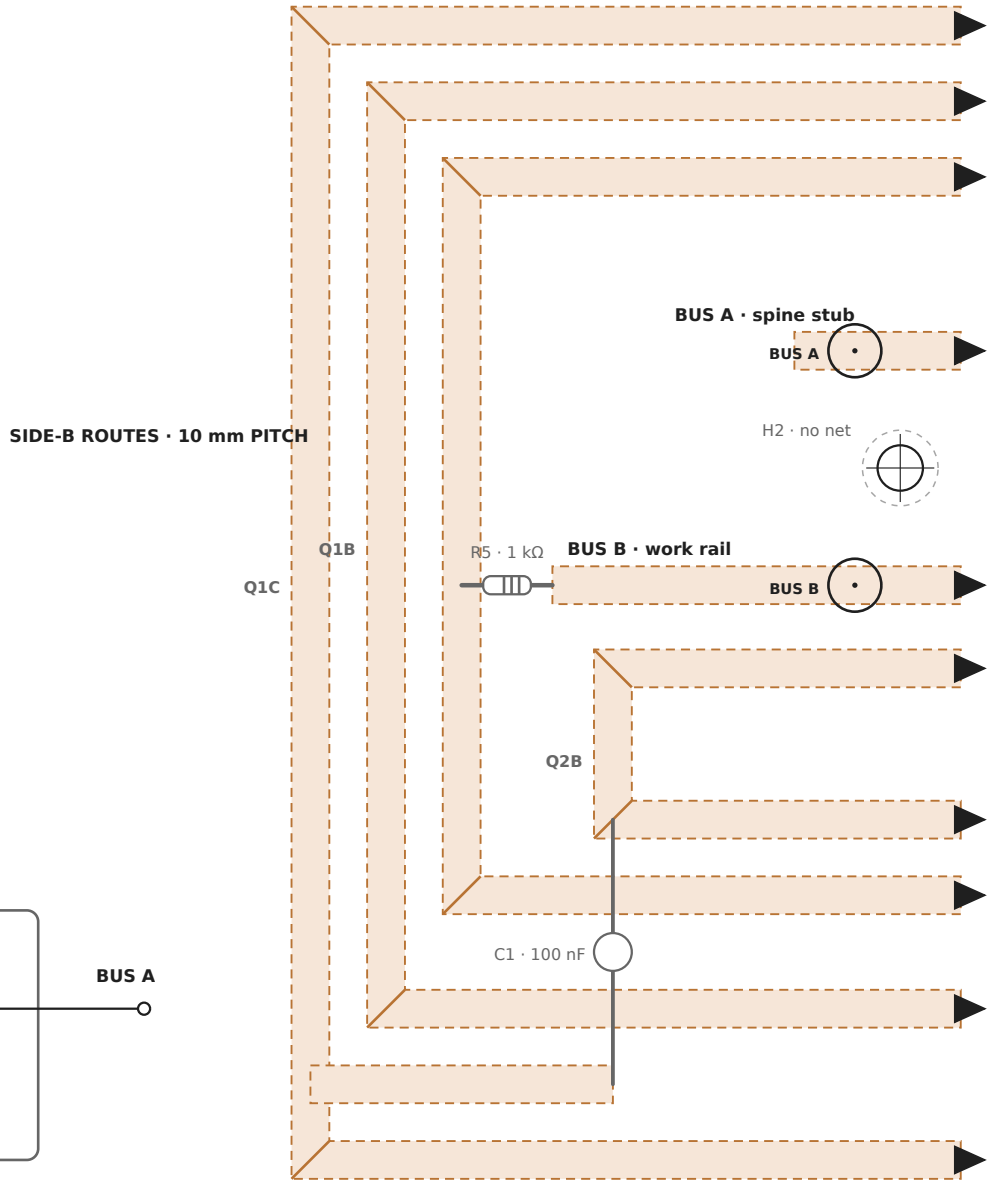
T6 Card 3: oscillator output

Four parallel back routes replace every insulated crossing on the oscillator face.

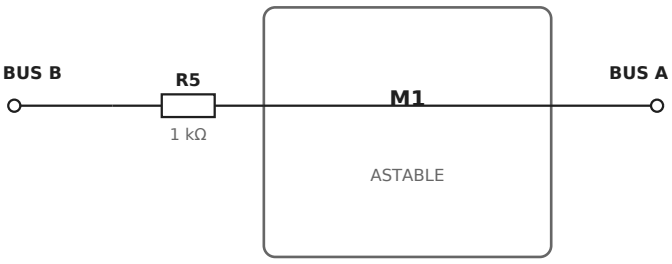


Side-B routes · protected BUS B

Match both ends of each Q1C, Q2B, Q1B and OUT route to side A. R5 connects only OUT to the shortened BUS B rail; the four channels must remain isolated.



Protected oscillator output



R5 limits a BUS B fault to less than 4.5 mA at 4.5 V.



Print at 100 %: this bar should measure 100 mm.

T6 Card 4: filter front

BUS A passes across side A; BUS B remains only a spine stub before reaching the back.



H1 · no net

Pass-through face

Card 4 does not consume BUS A, but carries it across its named work area. Wrap BUS B only at the spine so audio reaches the filter on side B.

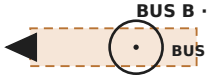
SPINE LINKS · TO PREVIOUS CARD

BUS A · work rail



H2 · no net

BUS B · spine stub



H3 · no net



Print at 100 %: this bar should measure 100 mm.

T6 Card 4: filter and output

BUS B feeds the M6 pencil filter and an integrated one-input M7 line-output stage.

V+ B

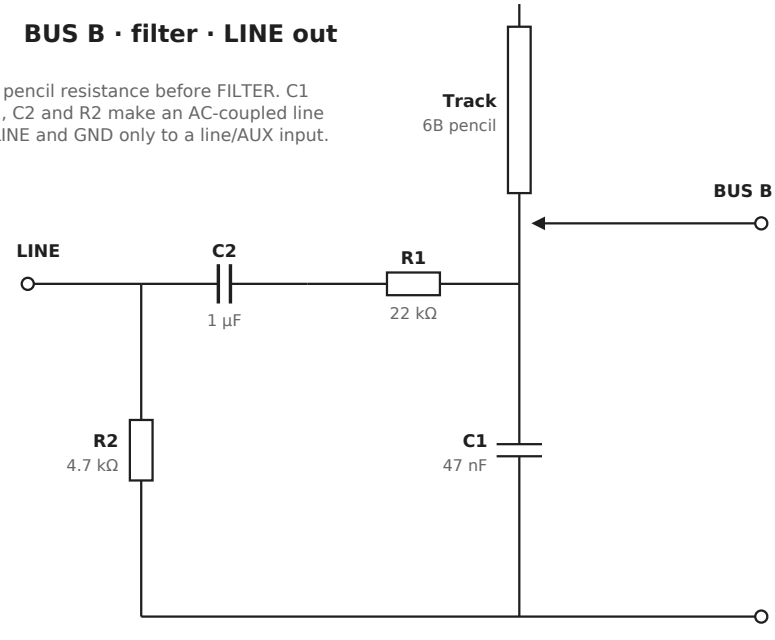
V+

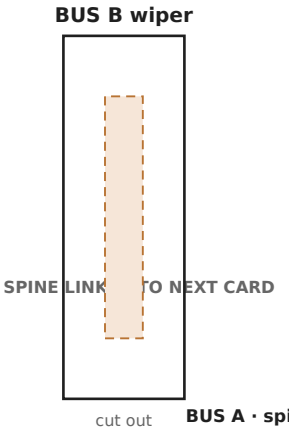
H1 · no net

Filter + line output schematic

BUS B · filter · LINE out

The wiper selects pencil resistance before FILTER. C1 removes buzz; R1, C2 and R2 make an AC-coupled line output. Connect LINE and GND only to a line/AUX input.





BUS A · spine stub

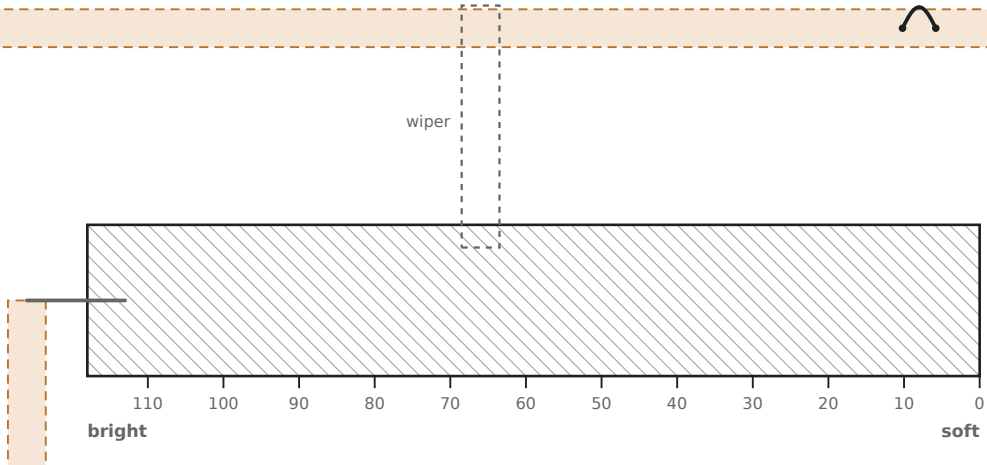
BUS A

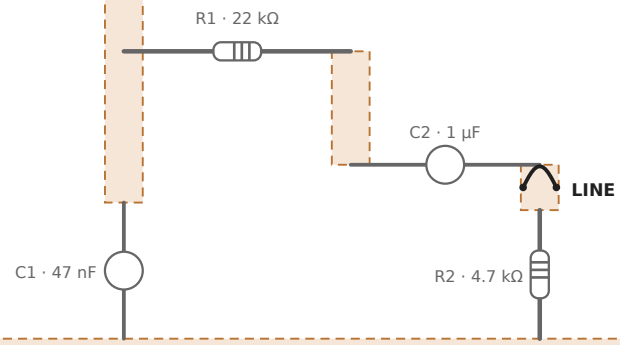
H2 · no net

BUS B · work rail

BUS B

BUS B





H3 · no net

GND B

GND

Print at 100 %: this bar should measure 100 mm.

T6 Build and test the four-card synth

Duplex-print pairs 1–2, 3–4, 5–6 and 7–8 at 100 %, flipping on the long edge. Print sheets 9 and 10 once.

H1 · no net

You need

- Four double-sided cards, a three-hole punch and a three-ring binder
- 5 mm copper tape for V+ and buses; 12 mm copper tape for GND
- Three sets of four flexible spine links made by the method selected in T4
- Card 1: switched 3×AA holder, 0.1 A resettable fuse, R1 1 kΩ and LED1
- Card 2: R1 13 kΩ, a soft 6B pencil, clear tape and card for one wiper
- Card 3: R1/R2/R5 1 kΩ, R3 10 kΩ, R4 4.7 kΩ, LED1 red, C1/C2 100 nF, Q1/Q2 2N3904
- Card 4: R1 22 kΩ, R2 4.7 kΩ, C1 47 nF, C2 1 µF, a soft 6B pencil, clear tape and card for one wiper
- Solder, clip leads, a multimeter, 3 AA cells, a line/AUX amp and a suitable cable

1 · Print the four cards

1. Duplex-print only the numbered pairs 1–2, 3–4, 5–6 and 7–8 at 100 %, flipping on the long edge.
2. Hold each card to a light. Its four spine wraps and link targets must register face-to-face, and no rail may enter a hole keep-out.
3. Punch and mark the cards 1 power, 2 controller, 3 oscillator and 4 filter/output.

2 · Build power and controller

1. On card 1, connect the switched holder through F1 to V+ and connect black to GND. Fit R1 and LED1; verify the switch and fuse are upstream of every other card.
2. On card 2, lay V+ and GND on both faces, BUS A on side A and BUS B on side B. Wrap each bus only to its opposite-face spine stub.
3. Build the pencil controller. With power off, verify all four link targets are isolated from unlike nets.

H2 · no net

3 · Build oscillator

1. On side A, stop BUS A at R4 and keep BUS B as a spine stub. Lay the eight labelled node ends without extending any across a bus.
2. On side B, join each matching pair through its own 10 mm-pitch channel. Fit R5 between OUT and the shortened BUS B rail; no channel may overlap another.
3. Check Q1C, Q2B, Q1B and OUT end-to-end through their reverse routes, then check every unlike pair for OL. Power the card and confirm LED1 flashes.

4 · Build filter and output

1. Lay the four rails and short opposite-face stubs. Keep BUS A as pass-through only.
2. Shade the tone track, fit C1 and the BUS B wiper, then add R1, C2 and R2 exactly as shown.
3. Verify LINE is AC-coupled and isolated from V+. Connect LINE and GND only to a powered speaker or amplifier line/AUX input—never headphones or a bare speaker.

5 · Link and test the book

1. Bind cards 1–2–3–4. With power off, join V+, BUS A, BUS B and GND at all three boundaries using the labelled targets.
2. Record each of the 12 link checks separately. Check every unlike pair for OL before inserting batteries.
3. Switch on and record BUS A and BUS B idle/active voltage, total current and LINE output amplitude.
4. Move the controller wiper to change pitch and the filter wiper to change tone. No clip-lead signal bypass is allowed.

6 · Flex and inspect

1. Open and close each page 100 times while the synth sounds. Record crackle, pitch jumps, silence or resets, then repeat at 1,000 cycles if it passes.
2. Remove BUS A at boundary 2–3: the oscillator must stop. Restore it and remove BUS B at boundary 3–4: LED1 must keep flashing while LINE goes quiet.
3. Recheck all 12 links and inspect every spine wrap for cracked copper, lifted adhesive or a shifted link.

Pass or fail

- **Pass:** four physical cards exchange power, pitch and audio only through labelled spine links; R5 protects BUS B; LINE drives a line input; flexing causes no dropout.
- **Fail:** a clip-lead bypass is needed, a bus crosses the wrong work area, BUS B can be hard-shorted by its driver, or any page requires a through-page via.

H3 · no net

T6 Four-card synth results

Record every inter-page link, bench voltage and current before and after flexing.

Boundary-by-boundary continuity

Measure with power disconnected. Every unlike pair must remain OL.

Boundary / net	New	After flex
1–2 · V+		
1–2 · BUS A		
1–2 · BUS B		
1–2 · GND		
2–3 · V+		
2–3 · BUS A		
2–3 · BUS B		
2–3 · GND		
3–4 · V+		
3–4 · BUS A		
3–4 · BUS B		
3–4 · GND		

Powered bench measurements

Bench measurement	Idle	Active
BUS A voltage		
BUS B voltage		
Total current		
LINE output amplitude		

Bus decision

Four physical cards were linked in the printed order	
Every boundary passed all four continuity checks	
BUS A occupied only side-A work areas	
BUS A controlled oscillator pitch across boundary 2–3	
BUS B occupied only side-B work areas	
R5 limited the BUS B short-circuit current	
BUS B carried audio across boundary 3–4	
LINE drove a line/AUX input without an external M7 page	

Notes

Pitch range, tone change and LINE level	
Idle/active current and BUS voltages	
Boundary crackle, dropout or flex damage	
Builder / date	

Print at 100 %: this bar should measure 100 mm.